

Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost &gt; 25/episode)



● SafeOT-typed (ours)    ▽ SDAC (multi-constraint SOTA)    ..... SAC-Pen coefficient sweep  
 ■ FDPI (constraint-based)    + SRCPO (multi-constraint SOTA)    --- FDPI  $p_f$  sweep  
 ◆ SAC-Lagrangian (penalty-based)    — typed Pareto frontier (soft  $\leq 25$ )    - - - SAC-Lag cost-limit sweep  
 ▲ SAC + fixed penalty (penalty-based)